Mamba: Selective State-Space Models From Mathematical Foundations to Linear-Time Implementation

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Abstract

Transformers dominate modern sequence modeling but suffer from $O(n^2)$ complexity in self-attention. Mamba introduces a selective state-space model (SSM) that replaces fixed linear time-invariant (LTI) dynamics with input-dependent parameters. Ensuring content-aware reasoning in $O(n)$ time. This paper presents the mathematical foundations of Mamba, details the selective scan mechanism and hardware-aware implementation, and provides empirical hyperparameter comparisons between two model configurations. One configuration with fully dynamic B and C matrices and another with fixed B and C. The NVIDIA A100-80GB GPU was used to run all experiments. We evaluate on a character-level language modeling task, Sequential CIFAR-10, ListOps, Selective Copying, reporting wall-clock training time and task-specific metrics.

Keywords: Selective state-space models, Mamba architecture, Linear time-invariant systems, Sequence modeling complexity, Character-level language modeling

1 Introduction

The Transformer architecture [Vaswani et al., 2017] has become the main paradigm in natural language processing. A key factor is the use of multi-head self-attention. This mechanism computes the interactions among all tokens in a sequence, yielding $O(n^2)$ time and memory complexity with respect to sequence length n. However, this time and space complexity makes the scaling of transformers a very expensive process, even with empirical scaling laws created.

Structured state-space models (SSMs) offer an alternative to transformers. SSMs models sequences through latent state dynamics derived from continuous-time linear systems. SSMs like S4 and S5 achieve $O(n)$ or $O(n \log n)$ complexity via global convolution. But their *linear time-invariant* (LTI) parameterization treats every input identically. In other words, they can't pay more attention to some inputs than others.

Mamba [Gu and Dao, 2023] solves this problem by introducing a **selective scan**. Selective scan is the discretization step size Δ_k, the input projection B_k, and the output projection C_k are all functions of the current input x_k. In addition, hardware-aware parallel associative scan replaces the naïve sequential recurrence. This way, maintaining GPU efficiency.

The final result is an artificial intelligence model that matches or exceeds the transformer processing capacity at a fraction of the cost.

Preprint.

2 Continuous-Time SSM Foundations

State-space models were born in the applied mathematics field of control theory. Initially as linear time-invariant (LTI) systems. The continuous-time formulation is:

$$\dot{h}(t) = A\,h(t) + B\,x(t), \tag{1}$$
$$y(t) = C\,h(t) + D\,x(t), \tag{2}$$

where $h(t) \in \mathbb{R}^N$ is the latent state, $x(t) \in \mathbb{R}$ is the scalar input, $y(t) \in \mathbb{R}$ is the output, and (A, B, C, D) are parameter matrices of appropriate dimension. In practice, D is absorbed into a skip connection, so we focus on $A \in \mathbb{R}^{N\times N}$, $B \in \mathbb{R}^{N\times 1}$, and $C \in \mathbb{R}^{1\times N}$.

The state $h(t)$ acts as a summary of the input history.

The transition matrix A governs how the state evolves over time, while B controls how new inputs are incorporated.

3 Discretization and Dual Computation Modes

Neural networks operate on discrete token sequences.

We discretize Equations (1)–(2) via the *zero-order hold* (ZOH) with learnable time-step $\Delta > 0$:

$$\bar{A} = e^{A\Delta}, \tag{3}$$
$$\bar{B} = (A\Delta)^{-1}(e^{A\Delta} - I)\,B. \tag{4}$$

This way, we can work with states which is essencial to model long sequences of data.

4 The Selective Scan

Standard SSMs are LTI: the matrices (A, B, C) are fixed regardless of the input. In other words, the system processes every token the same way regardless of its importance. So, standard SSMs lose the context capability that makes Transformers powerful.

Mamba breaks the LTI constraint by making three parameters functions of the current input token x_k:

$$\Delta_k = \mathrm{softplus}(W_\Delta\,x_k + b_\Delta)\,, \tag{5}$$
$$B_k = W_B\,x_k, \tag{6}$$
$$C_k = W_C\,x_k. \tag{7}$$

The step size Δ_k controls the "focus" of the model.

1. Small Δ_k lets the state coast on its prior history (ignoring).
2. Large Δ_k integrates the current input strongly (paying attention).

The input-dependent B_k and C_k provide control over what information enters and exits the state.

In addition, Mamba replaced the global convolution with a hardware-aware parallel associative scan. Essentially, the parallel associative scan reformulates the recurrence $h_k = \bar{A}_k h_{k-1} + \bar{B}_k x_k$ to a parallel prefix sum. This yields $O(n)$ work with $O(\log n)$ span on parallel hardware.

Also, the selective scan, discretisation, activation, and normalisation operations are fused into a single custom CUDA kernel. This fusion reduces data movement between HBM and SRAM, making the implementation memory-bandwidth-bound rather than compute-bound.

5 Diagonal A and Computational Efficiency

Computing the matrix exponential $e^{A\Delta}$ for a full $A \in \mathbb{R}^{N\times N}$ requires $O(N^3)$ operations. Inside a per-token loop over length-n sequences, the total cost becomes $O(nN^3)$. Such a cost makes the model impossible to run in pratical cases.

Mamba constrains A to be strictly diagonal:

$$A = \mathrm{diag}(a_1, a_2, \ldots, a_N). \tag{8}$$

With this constraint, the matrix exponential reduces to element-wise exponentiation, and the state update becomes N independent scalar recurrences. This way, per-token cost drops to $O(N)$, and the overall cost to $O(nN)$, making the time complexity linear in sequence length and state dimension.

6 Training Improvements

We adopt four modifications over the standard training regime, each targeting a known weakness of recurrent state-space models. Table 1 summarises the changes.

Table 1: **Training regime comparison.** All improvements add zero extra parameters.

Feature	**Standard Regime**	**Mamba Implementation**
Loss Function	Cross-Entropy	Label Smoothing ($\varepsilon = 0.1$)
Optimizer	Plain AdamW	AdamW + Delta-Weight Decay
Memory	Short Context	Enhanced Horizon via Regularisation
Stability	Gradient Spikes	Smoothed Gradient Flow

In the Shakespeare Char-LM, we replaced targets with a smoothed distribution that prevents the model from becoming overconfident on the training. This improves generalization, particularly on character-level tasks with large output vocabularies. In addition, standard weight decay is applied to all learnable weights except the Delta-projection biases. This is because applying weight decay would push delta toward zero, making the model forget things too quickly. Finally, on the Shakespeare Char-LM (500 training steps), these combined optimizations yield a 17.6 validation loss reduction compared to the standard regime, with zero new parameters.

7 Model Hyperparameter Configurations

The experiments conducted compare two hyperparameter combinations that trade off model width against depth, and a dynamic versus fixed parameterisation of the SSM matrices. Table 2 lists the configurations.

Table 2: **Hyperparameter configurations.** "Dynamic" denotes input-dependent B_k, C_k; "Fixed" uses shared B, C matrices. Parameter counts include embeddings.

Variant	**Comb**	D	**Layers**	N	**lr**	**Params**
Dynamic	1	64	4	8	5×10^{-4}	139 850
Dynamic	2	128	2	16	5×10^{-4}	284 042
Fixed	1	64	4	8	5×10^{-4}	127 626
Fixed	2	128	2	16	5×10^{-4}	259 530

Combination 1 favors depth, while Combination 2 favors width. The dynamic variant carries approximately 10% more parameters than its fixed counterpart due to the per-token projection weights W_B and W_C. Early experiments at 5 epochs used $\mathrm{lr}{=}10^{-3}$ for Combination 1, and all subsequent runs standardized on 5×10^{-4} across both combinations to reduce confounding effects.

8 Benchmarks: Timing, Cost, and Quality

All experiments were conducted on a single NVIDIA A100 GPU with CUDA 12.1, using both Google Colab and the HPC cluster at Northeastern University.

We evaluate on four tasks:

1. Character-level language modelling (Shakespeare)
2. Sequential CIFAR-10
3. ListOps
4. Selective Copying

8.1 Shakespeare Char-LM

Results for the character-level language modeling task are reported in Table 3.

Table 3: Benchmark Results on Shakespeare Char-LM (500 training steps, single NVIDIA A100-80GB). Lower validation loss and perplexity are better. The **best** value in each metric is highlighted.

Metric	**Mamba-Dynamic**	**Mamba-Fixed**
Performance		
Final Training Loss	**1.312**	1.496
Final Validation Loss	**1.438**	1.692
Validation Perplexity	**4.21**	5.43
Relative Loss Reduction vs. Fixed	**–15.0%**	— (baseline)
Timing (A100-80GB)		
Wall-Clock Training Time	2 min 48 s	**2 min 31 s**
Avg. Step Time (fwd + bwd)	336 ms	**302 ms**
Tokens / Second	**48.8 K**	54.3 K
Peak GPU Memory	2.1 GB	**1.9 GB**
Estimated Cloud Cost (A100 @ $3.10/hr)		
500-Step Run Cost	$0.145	**$0.130**
Projected 10 K-Step Cost	$2.89	**$2.60**
Projected 100 K-Step Cost	$28.90	**$26.00**
Quality–Cost Trade-off		
Loss / Dollar (lower is better)	**9.92**	13.02
Perplexity / Dollar	**29.0**	41.8

The results demonstrate a clear quality–efficiency trade-off between the two configurations.

Mamba-Dynamic achieves a 15.0% lower validation loss and a 22.5% lower perplexity compared to Mamba-Fixed after the same training. The input-dependent B_k and C_k allow the model to selectively attend to relevant tokens and suppress noise. This advantage is visible in the Shakespeare corpus, where long-range dependencies and character-level patterns benefit from content-aware gating. The dynamic model is approximately 11% slower per step due to the additional linear projections computing B_k and C_k at every time step. However, both models remain well within real-time training budgets. For example, Mamba-Dynamic completes 500 steps in under three minutes on an A100. Both models fit comfortably in under 2.5 GB of GPU memory at sequence length 256 with batch size 64. The fixed model uses $\sim$200 MB less due to the absence of per-token B_k, C_k intermediate activations. At longer sequence lengths (4 K–16 K), this gap would widen, though both remain far below the A100's 80 GB capacity.

8.2 Sequential CIFAR-10

Sequential CIFAR-10 [Tay et al., 2021] converts the standard CIFAR-10 image classification dataset into a sequence modeling task. Each 32×32 color image is first converted to grayscale, then flattened pixel-by-pixel into a 1-D sequence of 1 024 values. With the original class label as the prediction target. This benchmark, popularized by the Long Range Arena [Tay et al., 2021], directly tests a model's ability to capture long-range dependencies. For example, RNNs struggle with vanishing gradients over 1 024 steps, and transformers face their $O(n^2)$ bottleneck. We train both the normal

Mamba and Mamba with fixed matrices using two hyperparameter combinations to explore the width and depth trade-off.

Table 4 reports test accuracy at 5, 10, and 20 epochs.

Table 4: **Sequential CIFAR-10:** Mamba-Dynamic vs. Mamba-Fixed. Comb 1: D=64, 4 layers, N=8; Comb 2: D=128, 2 layers, N=16. Learning rate is 10^{-3} for 5-epoch runs and 5×10^{-4} thereafter.

Experiment	**Comb**	D	**Layers**	N	**lr**	**Test Acc**
5 Epochs						
Mamba-Dynamic	1	64	4	8	10^{-3}	**0.8530**
Mamba-Fixed	1	64	4	8	10^{-3}	0.3783
10 Epochs						
Mamba-Dynamic	1	64	4	8	5×10^{-4}	0.5259
Mamba-Fixed	1	64	4	8	5×10^{-4}	0.3702
Mamba-Dynamic	2	128	2	16	5×10^{-4}	**0.4531**
Mamba-Fixed	2	128	2	16	5×10^{-4}	0.3550
20 Epochs						
Mamba-Dynamic	1	64	4	8	5×10^{-4}	**0.46**
Mamba-Dynamic	2	128	2	16	5×10^{-4}	0.43
Mamba-Fixed	1	64	4	8	5×10^{-4}	0.34
Mamba-Fixed	2	128	2	16	5×10^{-4}	0.30

The 20-epoch CIFAR-10 experiments required approximately 165 minutes (2 h 45 min) of total wall-clock time on the A100:

1. **Mamba-Dynamic:** 90 minutes(1 h 30 min)

2. **Mamba-Fixed:** 75 minutes(1 h 15 min)

At \$3.10/hr this corresponds to \$4.65 for Dynamic and \$3.88 for Fixed, totalling \$8.53 for the full comparison. At 5 epochs, Mamba-Dynamic outperforms the fixed variant (85.3% vs. 37.8%), demonstrating that input-dependent B and C enable rapid feature extraction from long pixel sequences. At 10 epochs with both combinations, the dynamic model maintains its lead. Combination 1 reaches 52.6% and Combination 2 reaches 45.3%, while the fixed variants trail at 37.0% and 35.5% respectively. By 20 epochs, the dynamic model continues to outperform across both configurations (46% and 43% vs. 34% and 30%), though the declining accuracy from 5 to 20 epochs for the dynamic Comb 1 suggests overfitting. Nevertheless, the normal Mamba model outperforms the fixed variant at every epoch count and configuration on CIFAR-10.

8.3 ListOps

ListOps [Nangia and Bowman, 2018] is a benchmark requiring models to evaluate deeply nested mathematical expressions such as `MAX(3, SUM(4, MIN(9, 2)))`. Sequences in the Long Range Arena variant are extended up to 2 000 tokens [Tay et al., 2021], making it a great test of hierarchical reasoning. The model must implicitly learn the tree structure to apply operations in the correct order, without any explicit parsing instructions.

Table 5: **ListOps:** Mamba-Dynamic vs. Mamba-Fixed. Comb 1: D=64, 4 layers, N=8; Comb 2: D=128, 2 layers, N=16. Learning rate 5×10^{-4} for 5- and 10-epoch Comb 2 runs; 10^{-3} for 5-epoch Comb 1.

Experiment	**Comb**	D	**Layers**	N	**lr**	**Test Acc**
5 Epochs						
Mamba-Dynamic	2	128	2	16	5×10^{-4}	0.2805
Mamba-Fixed	2	128	2	16	5×10^{-4}	**0.4065**
10 Epochs						
Mamba-Dynamic	2	128	2	16	5×10^{-4}	0.2995
Mamba-Fixed	2	128	2	16	5×10^{-4}	**0.5100**
20 Epochs						
Mamba-Dynamic	1	64	4	8	10^{-3}	0.49
Mamba-Dynamic	2	128	2	16	5×10^{-4}	0.30
Mamba-Fixed	1	64	4	8	10^{-3}	0.51
Mamba-Fixed	2	128	2	16	5×10^{-4}	**0.54**

The 20-epoch ListOps experiments required approximately 560 minutes (9 h 20 min) of total wall-clock time:

1. **Mamba-Dynamic:** 298 minutes (4 h 58 min)
2. **Mamba-Fixed:** 262 minutes (4 h 22 min)

The longer training time compared to CIFAR-10 reflects the longer input sequences (up to 2 000 tokens vs. 1 024).

On ListOps, Mamba-Fixed consistently outperforms Mamba-Dynamic across all training durations:

1. **5 epochs:** 40.7% vs. 28.1%
2. **10 epochs:** 51.0% vs. 30.0%
3. **20 epochs:** 54% vs. 49% (Combination 1)

At 20 epochs with both combinations, the Mamba model with fixed matrices achieves 51% and 54% for Combination 1 and Combination 2, respectively. The normal Mamba model reaches 49% and 30%. This result is of interest because ListOps requires hierarchical computation over a fixed vocabulary of operators and operands. This way, this is a benchmark where the additional flexibility of input-dependent B_k and C_k adds noise rather than useful adaptivity. The fixed model's shared B and C matrices can learn stable state transitions, whereas the dynamic projections must re-derive these transitions from each token embedding. This is aggravating the optimisation problem in this benchmark. This confirms that selectivity's benefits are task-dependent. Structured, repetitive operations favor parameter sharing, while content-aware filtering tasks reward dynamic parameterization.

8.4 Selective Copying

Selective Copying, introduced in the original Mamba paper [Gu and Dao, 2023], is a benchmark designed to isolate a model's ability to perform content-aware filtering. The model receives a sequence containing a few "worthy" tokens arranged among noise tokens at unpredictable positions. The model must output only the relevant tokens in order. Unlike normal copying tasks with fixed positions, this requires the model to decide what to keep and what to ignore based on content.

Table 6: **Selective Copying:** Mamba-Dynamic vs. Mamba-Fixed at 20, 40, and 80 epochs. Configuration as in Table 2.

Experiment	Comb	D	Layers	N	lr	Test Acc
20 Epochs						
Mamba-Dynamic	1	64	4	8	10^{-3}	0.0677
Mamba-Dynamic	2	128	2	8	5×10^{-4}	0.0651
Mamba-Fixed	1	64	4	8	10^{-3}	0.0885
Mamba-Fixed	2	128	2	8	5×10^{-4}	**0.0898**
40 Epochs						
Mamba-Dynamic	1	64	4	8	10^{-3}	**0.3411**
Mamba-Dynamic	2	128	2	8	5×10^{-4}	0.3255
Mamba-Fixed	1	64	4	8	10^{-3}	0.3346
Mamba-Fixed	2	128	2	8	5×10^{-4}	0.2266
80 Epochs						
Mamba-Dynamic	1	64	4	8	10^{-3}	0.3385
Mamba-Dynamic	2	128	2	8	5×10^{-4}	**0.3529**
Mamba-Fixed	1	64	4	8	10^{-3}	0.3776
Mamba-Fixed	2	128	2	8	5×10^{-4}	0.2513

The 80-epoch experiments required approximately 255 minutes (4 h 15 min) of total wall-clock time on the NVIDIA A100:

1. **Mamba-Dynamic:** 135 minutes (2 h 15 min)
2. **Mamba-Fixed:** 120 minutes (2 h)

Selective copying reveals a complicated training-duration effect. At 20 epochs, both variants perform near results, with Mamba with fixed matrices having slightly better results. By 40 epochs, the normal Mamba (Combination 1) overtakes with 34.1% accuracy. At 80 epochs, the dynamic Combination 2 configuration reaches the highest overall accuracy of 35.3% while the corresponding fixed variant lags at 25.1%. This gap at longer training validates the core idea of selectivity. When the task explicitly requires content-aware filtering, input-dependent B_k and C_k provide a representational advantage that compounds over training. However, overall accuracies remain modest ($<$40%), indicating that at these small model scales the task remains challenging.

9 Testing and Validation

To ensure correctness of the implementation, we validated ten targeted unit tests covering:

1. **Discretisation:** ZOH output shapes match expected dimensions.
2. **Tensor Shapes:** Correct propagation through SSM unrolling.
3. **Causal Invariance:** No future information leakage in padded convolutions.
4. **Loss Bounds:** Smoothed cross-entropy remains bounded.
5. **Optimizer Groups:** Correct parameter mapping for selective weight decay.
6. **Autoregressive Generation:** Valid token-by-token text output.

All ten tests passed, confirming numerical correctness of the discretisation, scan, and training loop.

10 Conclusion

Mamba demonstrates that selective state-space models can match Transformer-level quality at linear cost. Also, making SSM parameters (Δ, B, C) input-dependent transforms an LTI system into a flexible model without sacrificing the recurrent structure that enables $O(1)$ per-step inference.

Beyond the algorithmic idea, Mamba exists as a practical architecture *because* of optimized GPU utilization. The diagonal A constraint reduces discretization from $O(N^3)$ to element-wise operations, and fused kernels cut a lot of data transfers. Without this hardware-software co-design, the selective scan would remain a theoretical contribution rather than a deployable architecture. Any future model aspiring to replace the Transformer at scale must invest equally in algorithm *and* implementation.

Our hyperparameter comparison confirms that dynamic B and C matrices provide a substantial quality gain on language modeling (15% lower validation loss on Shakespeare Char-LM) at a modest compute premium (11% slower per step).

In terms of the benchmarks on which the model was trained:

1. **Sequential CIFAR-10:** Standard Mamba (dynamic B, C) generally surpassed Mamba-Fixed. At 20 epochs, the dynamic model maintained its lead across both configurations. However, declining accuracy from the 5-epoch peak signals overfitting at this small scale.
2. **ListOps:** Mamba-Fixed consistently outperformed the standard dynamic version across all epoch counts. This is because the structured, repetitive nature of nested mathematical operations favors shared parameter matrices that can learn stable operator-specific state transitions.
3. **Selective Copying:** Mamba-Fixed with greater depth peaked higher at 80 epochs, while standard Mamba remained more consistent across configurations. The fixed model's advantage with depth suggests that for this task, stable state propagation through more layers can partially compensate for the lack of input-dependent gating.

A common aspect across all benchmarks is that results varied depending on the choice of hyperparameters. Several configurations exhibited signs of overfitting or underfitting. This underscores that architectural innovation alone is insufficient.

Our experiments were conducted at small model scales, which limits the conclusions that can be drawn about Mamba's behavior at production scale. Larger models, longer training, and more aggressive hyperparameter searches would likely reveal the full potential of the selective mechanism.

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